

MEVIN JOSE

mjenius1357@gmail.com | +91 9902932228 | Bangalore, India
linkedin.com/in/mevin-jose | mjenius.github.io/Portfolio | github.com/mjenius

SUMMARY

AI/ML undergraduate specializing in agentic systems, explainable RL, and LLM-augmented pipelines. Built a self-correcting SQL agent featuring a multi-agent (planner-executor-evaluator) architecture yielding 95–100% query success, a PPO traffic agent cutting emergency transit time by 10.6%, and a real-time CV system sustaining <100ms latency on 720p feeds. Focus on evaluation rigor, SHAP-based interpretability, and validated on real-world datasets and controlled benchmarks.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java

ML & AI: LightGBM, Scikit-learn, PyTorch, Stable-Baselines3, SHAP, OpenCV, NLP

Agentic AI: Groq/Gemini APIs, FAISS, Sentence Transformers, RAG, Llama 3, Prompt Engineering

Databases: PostgreSQL, MongoDB, SQLite, Redis, Pinecone, Vector Databases

Systems & MLOps: FastAPI, Docker, CI/CD, GitHub Actions, Drift Detection, Model Monitoring

Frameworks: Streamlit, Flask, Pandas, NumPy, SciPy, Socket.IO

EXPERIENCE

Superhero Learning

Bangalore, India

Web Development Intern

Jun – Sep 2025

- Built and deployed 8+ Next.js production pages serving 100+ active users; optimized MongoDB indexing and query plans, reducing API latency by 25%.
- Designed performance observability and asset compression workflows, improving load speed by 10% and reducing iteration time by 25%.

PROJECTS

AI Data Analyst Agent

Python, FastAPI, FAISS, Sentence Transformers, Groq/Gemini APIs, RAG, Pydantic

- Enabled automated analytics workflows, reducing manual SQL effort via a multi-agent (planner-executor-evaluator) architecture with automated self-correction using database feedback, achieving 95–100% query success on Olist e-commerce dataset.
- Built semantic RAG layer (FAISS + Sentence Transformers) over relational schema; implemented multi-model API fallback cascade (Groq → Gemini) with deterministic SQL fallback; streamed reasoning traces via SSE.

Football Recruitment & Scouting Decision Support System

Scikit-learn, SHAP, Llama 3 (Ollama), FastAPI, Docker, Streamlit

- Built clustering and similarity search pipeline over 3,364 player records with market valuation modeling ($R^2 = 0.82$); applied SHAP for interpretability and Llama 3 for automated scouting report generation, reducing manual shortlisting time by ~70%.
- Deployed Dockerized FastAPI inference with reproducible pipelines and Streamlit dashboard for stakeholder-facing analytics end-to-end ML pipeline.

AEGIS - Tamper-Resistant Surveillance System

OpenCV, HMAC, Socket.IO, Flask, FFmpeg | Top-10 Kodikon

- Engineered real-time CV pipeline detecting blur, glare, and replay attacks; sustained <100ms latency on 720p feeds using optimized OpenCV processing with scalable ingestion-to-detection architecture.
- Implemented HMAC-based watermark validation and Socket.IO dashboard for real-time tamper alerts across live video streams.

Adaptive Traffic-Signal Control with Explainable AI

PPO, SUMO, Stable-Baselines3, SHAP

- Trained PPO agent reducing emergency vehicle transit time by 10.6% vs fixed-timing baseline; validated robustness across controlled simulation stress scenarios.
- Applied SHAP to interpret 43D state space and designed reward decomposition for policy reliability in safety-critical edge cases.

Demand Forecasting ML System

LightGBM, FastAPI, Pandas, SciPy, SQLite, Parquet

- Feature-engineered pipeline over 30K+ SKUs (~2M+ records); benchmarked LightGBM against statistical baselines achieving 25% MAPE reduction via time-aware cross-validation; implemented rolling-window drift monitoring.

EDUCATION

PES University

Bangalore, India

B.Tech, Computer Science (AI & Machine Learning)

Aug 2023 – May 2027

CERTIFICATIONS

Google Data Analytics | Meta GenAI | Microsoft Computer Vision | Kaggle ML & DL